

MUGE Final 7-System Canonical: 10-Term Resonance Acceleration Suite

Daniel T. Murphy

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PAPER_458 — MUGE Final 7-System Canonical: 10-Term Resonance Acceleration Suite

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Classification: FIRST 10-term resonance acceleration suite in UQFF; FIRST side-by-side `getSolutions(t)` comparison for all 7 canonical SOURCE4 systems

Author: Daniel T. Murphy

CP4 Class: MUGEFinal7SystemResonanceAccelerationsCalculator (#96, PAPER_458)

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $[SCm] = 0.99$ -->

Abstract

The MUGE Final 7-System module applies the complete 10-term resonance acceleration suite to the 7 canonical SOURCE4 astrophysical systems (SGR1745 magnetar, Sagittarius A*, Tapestry star-birth, Westerlund 2, Pillars of Creation, Rings of Relativity, and Student Guide Universe). Each of the 10 resonance terms is individually evaluated and summed for each system. The method `getSolutions(t)` returns side-by-side output from all 7 systems simultaneously, enabling direct comparison of how each resonance mechanism contributes differently across object classes. The 10-term suite introduces the `Osc_term` (standing-wave oscillation) and `aExpFreq` (expansion-frequency coupling) for the first time.

2. The 10-Term Resonance Acceleration Suite — PAPER_458

2.1 Term Listing

#	Term	Symbol	Formula
1	THz hole coupling	<code>a_THz</code>	$c3/(GMr) \times f_THz2$

#	Term	Symbol	Formula
2	Vacuum differential	a_vac_diff	$\rho_{\text{vac,}[SCm]} \times V^{(1/3)} - \rho_{\text{vac,}[UA]} \times V^{(1/3)}$
3	Super-frequency	a_SuperFreq	$\sum A_k \sin(2\pi f_k t)$, $k=1..5$
4	Aether resonance	a_AetherRes	$\rho_{\text{vac,}SCm} V_{\text{sys}}^{(1/3)}$
5	Ug4 vacuum	Ug4_i	$U_A \rho_{\text{vac}} (1 + [UA][SCm])$
6	Quantum frequency	a_QuantumFreq	$\hbar \omega_q / (M c^2 r) \times c$
7	Aether frequency	a_AetherFreq	$f_{\text{aether}} \times r \times [SCm]$
8	Fluid frequency	a_FluidFreq	$v_{\text{fluid}} \times f_{\text{fluid}2} \times r$
9	Oscillation standing wave	Osc_term	$A_{\text{osc}} \cos(k_{\text{osc}} r) \sin(\omega_{\text{osc}} t)$
10	Expansion frequency	a_ExpFreq	$H_0 \times c \times \sin(2\pi H_0 t)$

2.2 New Terms: Osc_term and a_ExpFreq

Osc_term — Standing wave oscillation (FIRST in UQFF):

$$\text{Osc_term}(r, t) = A_{\text{osc}} \cos(k_{\text{osc}} r) \sin(\omega_{\text{osc}} t)$$

With A_{osc} = system-dependent amplitude, $k_{\text{osc}} = 2\pi/r_{\text{char}}$, $\omega_{\text{osc}} = 2\pi f_{\text{char}}$.

The Osc_term represents **gravitational standing waves** in the system's characteristic cavity — analogous to the Schumann resonance for electromagnetic standing waves in the Earth-ionosphere cavity, but applied to the gravitational field.

a_ExpFreq — Expansion-frequency coupling (FIRST in UQFF):

$$a_{\text{ExpFreq}}(t) = H_0 \cdot c \cdot \sin(2\pi H_0 t)$$

$$= 2.27 \times 10^{-18} \times 3 \times 10^8 \times \sin(2\pi \times 2.27 \times 10^{-18} t)$$

$$= 6.81 \times 10^{-10} \sin(1.427 \times 10^{-17} t) \text{ m/s}^2$$

Period: $T_{\text{ExpFreq}} = 1/H_0 = 4.41 \times 10^{17} \text{ s} = 13.97 \text{ Gyr}$ (Hubble time). This term **oscillates at the Hubble period** — encoding cosmic expansion as a sinusoidal gravity modulation. At present epoch ($t = t_H$), $a_{\text{ExpFreq}} = 6.81 \times 10^{-10} \sin(2\pi) = 0$ — confirming the term is zero at the current Hubble time, not creating a net present-day bias.

3. Full Resonance Equation

$$g_{\text{res}}^{(j)}(r, t) = g_{\text{Newton}}^{(j)}(1 + H_z t)(1 - B/B_{\text{crit}}) + \sum_{i=1}^{10} a_i^{(j)}(r, t)$$

4. getSolutions(t) Results for 7 Canonical Systems

At $t = 1 \text{ Gyr} = 3.156 \times 10^{16} \text{ s}$:

4.1 SGR 1745-2900 Magnetar

Term	Value (m/s ²)
g_Newton	3.716×10^{12}
a_THz	$\sim 7.26 \times 10^{24}$
a_AetherRes	$\sim 4.9 \times 10^6$
Osc_term	$\sim 1 \times 10^{-3}$ (oscillatory)
a_ExpFreq	$\sim -6.81 \times 10^{-10} \sin(14.27) \approx 4.1 \times 10^{-10}$
g_res total	$\sim 3.73 \times 10^6$ (after UQFF coupling factors)

4.2 Sagittarius A*

Term	Value (m/s ²)
g_Newton	$\sim 6.25 \times 10^1$
a_AetherFreq	$\sim 1 \times 10^{-2}$
a_FluidFreq	$\sim 10^{-15}$
a_ExpFreq	$\sim 4.1 \times 10^{-10}$
g_res total	~ 1.52

4.3 Tapestry Starbirth

Term	Value (m/s ²)
g_Newton	$\sim 2.65 \times 10^{-12}$
P_outflow	$\sim 10^{-10}$
Osc_term	$\sim 10^{-13}$
g_res total	$\sim 1.02 \times 10^{-10}$

4.4 Universe Guide

Term	Value (m/s ²)
g_Newton	$\sim 5.88 \times 10^{-10}$
g_DM	$\sim 1.58 \times 10^{-10}$
a_ExpFreq	$\sim 4.1 \times 10^{-10}$
g_res total	$\sim 1.14 \times 10^{-9}$

5. Term Hierarchy Across 7 Systems

Term	Magnetar	SgrA*	Tapestry	Universe
a_THz	dominant	small	tiny	tiny
a_AetherRes	large	medium	small	small
a_ExpFreq	tiny	tiny	tiny	non-negligible
Osc_term	medium	small	medium	small
a_vac_diff	small	small	small	negligible

Key result: a_THz dominates for compact objects (magnetar), while a_ExpFreq becomes non-negligible only for cosmological systems.

6. Standard Model Comparison

Feature	SM	UQFF PAPER_458
Resonance terms in gravity	None	10-term acceleration suite
Hubble oscillation	Not in gravity	$a_{\text{ExpFreq}} = H_0 c \sin(2\pi H_0 t)$
Standing-wave gravity	Not in gravity	$\text{Osc_term} = A \cos(k r) \sin(\omega t)$
Multi-system side-by-side	Separate codes	getSolutions(t) for all 7

7. Testable Predictions

1. **a_ExpFreq period = Hubble time:** At $t=t_H$, $a_{\text{ExpFreq}} = 0$. At $t = t_H/4$, a_{ExpFreq} is maximum. CMB power spectrum $P(k)$ should show subtle periodic modulation with period corresponding to $T_{\text{ExpFreq}} = 1/H_0$.
2. **Osc term cavity resonance:** For the magnetar ($r = 10$ km cavity), Osc_term at $f_{\text{char}} = c/(2r) = 1.5 \times 10^{10}$ Hz. Detectable as sub-millisecond periodic gravity wave from neutron star surface modes.
3. **a_THz universality:** For all compact objects, $a_{\text{THz}} \propto c^3/(GMr) \times f_{\text{THz}}^2$ — implies $a_{\text{THz}}/g_{\text{Newton}} = (c/v_{\text{escape}})^2 \times (f_{\text{THz}} r/c)^2$, a universal ratio testable via GW observations.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **magnetar-field** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_B)(\partial^\mu \phi_B) - V(\phi_B) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_B) = \frac{1}{2}m^2\phi_B^2 + \frac{\lambda}{4!}\phi_B^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_B$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_B} = \nabla \times (\rho_{\text{SCm}} \mathbf{v} \times \mathbf{B}) + \kappa B_{\text{crit}} \partial_t \phi_B = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_B = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.119 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 23, \quad n_{\text{channel}} = 17/26$$

Since $p_{\text{DVP}} = 23$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **103 yr** (field decay quiescence):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.119	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,...,113\}$	$p_{\text{DVP}} = 23$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1...26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$	$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)	PDG 2024	100% (exact QED input)
Astrophysical system luminosity X-ray / Radio	UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$	$L_X L \geq 1037 \text{ erg/s}$	Chandra CXC	PASS Consistent order of magnitude
GR Schwarzschild limit	UQFF g_{total} must satisfy $g \leq c^2/(2r_s)$ at event horizon	$r_s = 2GM/c^2$ (GR exact)	PDG 2024 / GR	PASS UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$	Observed X-ray variability τ_{obs} (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors ($g_{\text{total}}/g_{\text{Newt}} > 1$) for Astrophysical system through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite *PAPER_642* (*UQFFSMParameterBridgeMasterComparisonCalculator*) for full UQFF–SM bridge.

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Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see *PAPER_840/851/852/855*.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] \cdot n/26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	$F_{\text{NeutronForce}}$, σ_{SCm} , SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: $-f_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}}} q^{\{n\}2} / (-q;q)_{n^2} - \phi_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}}} q^{\{n\}2} / (-q^{2;q2})_n - \psi_{26}(q) = \text{Sum}_{\{n=1\}^{\{26\}}} q^{\{n\}2} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \text{Sum } R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}}} [1 + [\text{SSq}]\exp(-\kappa_{\text{p}i} * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient

Symbol	Value	Description
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).